

1. Data Generalizability and Industrial Application (Malhotra Gap 1)

The core concern here is the external validity of academic research. Research consistently finds that models trained and tested exclusively on public, open-source datasets (like NASA and PROMISE) do not reliably reflect the characteristics or complexity of proprietary industrial systems. The lack of access to industrial data means that claims of high predictive performance are rarely verified under real-world, closed-source constraints. This gap necessitates an effort to either secure and utilize proprietary datasets for external validation or, failing that, develop novel techniques capable of simulating the unique characteristics of industrial software development data.

2. Multi-Label Actionable Intelligence (Pachouly Gap 1)

A primary criticism of current SDP models, despite their high binary classification accuracy, is their low practical utility. Simply classifying a module as "buggy" or "non-buggy" does not provide the actionable intelligence required by developers to prioritize and fix defects. This research void demands extending the predictive scope to **Multi-Label Classification (MLC)**. Future models must predict secondary, actionable attributes, such as defect severity (High, Medium, Low), the estimated effort/cost to fix (e.g., using Story Points), and information regarding which resource should be allocated. Addressing this requires a fundamental change in the data source, as existing public metric sets lack the necessary labeled ground truth for these attributes.

3. Automatic Feature Extraction and Artifact Fusion (Giray Gap 1 & 2)

Deep Learning promises automatic feature engineering, circumventing manual metric selection. However, many current DL models still rely on basic, hand-crafted metrics. The path forward requires fulfilling DL's original promise: **developing more comprehensive DL architectures that automatically capture sophisticated semantic features** directly from raw code structures (ASTs, CFGs). Furthermore, a significant amount of semantic information about defects is contained not just in the code but in external artifacts like bug reports, requirements, and commit messages. Future work must address the methodological challenge of **fusing these diverse, heterogeneous software artifacts** (code, text, numerical metrics) into a unified representation to create a richer predictive signal.

4. Model Optimization and Hyperparameter Tuning (Zain Gap 1)

The complexity of modern DL models means that their performance is acutely sensitive to their architectural and hyperparameter configuration (e.g., number of layers, learning rate, and activation functions). Determining the optimal configuration through manual trial and error is infeasible. Therefore, a critical technical gap lies in the **optimization of these performance-influencing parameters** using advanced, often search-based, optimization techniques. This systematic tuning is vital to ensure that the prediction results achieved are not merely stochastic artifacts but represent the true predictive capability of the DL architecture.

5. Multi-Objective Optimization (Malhotra Gap 2)

Most SDP studies define "optimal" solely in terms of maximum predictive performance (e.g., highest AUC). This single-objective focus ignores critical constraints necessary for real-world deployment. Deployable models must also satisfy practical requirements like low computational cost, fast execution time, and model interpretability. This necessitates the development and evaluation of **multi-objective optimization (MOO) approaches**. MOO allows

researchers to simultaneously seek a balance across a Pareto front of conflicting objectives, such as maximizing AUC while minimizing computational training time. Currently, only about 11% of hybrid studies utilize MOO, confirming this as a significant methodological void.

6. Standardization of Evaluation Metrics (Zain Gap 2)

The inconsistency and inappropriateness of evaluation metrics remain a technical liability. Over-reliance on threshold-dependent metrics like Accuracy and Recall in heavily imbalanced SDP datasets compromises the reliability of conclusions and prevents robust meta-analysis. This gap mandates the **standardization of evaluation metrics**, emphasizing the universal adoption of stable measures such as AUC (Area Under the Curve) and comprehensive confusion matrices. This standardization is crucial for ensuring that cross-study comparisons are statistically valid and that the reported predictive capacity accurately reflects a model's performance on minority classes.

7. Explicit Data Validation and Quality Assurance (Pachouly Gap 2)

Despite the reliance on historical data, early SDP research frequently neglects robust data quality assessment. The high stakes involved in defect prediction—where incorrect classification can lead to skipped testing of critical code—require high data integrity. This gap requires future research to **incorporate robust, explicit data validation and quality assessment methods** prior to model training. Validation techniques, including assessing internal data correlations, outlier removal (noise handling), and formal statistical tests (Parametric/Non-Parametric) for feature richness, must become a mandatory step to certify the dataset's appropriateness for the intended complex predictive task (like MLC).